

Question: 1

Given the network 192.168.10.0/25, calculate the subnet mask, network address, broadcast address, and the range of usable IP addresses.

Answer: 1

**Subnet Calculation Breakdown (192.168.10.0/25)**

To solve the network parameters for a /25 CIDR prefix, we break down the 32 bits of an IPv4 address. A /25 means the first 25 bits belong to the network portion, leaving exactly **7 bits for the host portion** (32–25=7).

- **1. Subnet Mask:**
  - In binary, the first 25 bits are set to 1 and the remaining 7 bits are 0:  
11111111.11111111.11111111.10000000
  - Converting this binary string back to standard decimal gives: **255.255.255.128**
- **2. Network Address:**
  - This is the very first address of the subnet block, where all host bits are set to 0.
  - Value: **192.168.10.0**
- **3. Broadcast Address:**
  - This is the very last address of the subnet block, where all host bits are set to 1.
  - The size of the subnet block is  $2^7=128$  absolute addresses (spanning from 0 to 127).
  - Value: **192.168.10.127**
- **4. Range of Usable IP Addresses:**
  - Usable IPs sit strictly between the Network ID and the Broadcast Address.
  - Range: **192.168.10.1 to 192.168.10.126** *(Total of 126 usable host addresses)*

| Final Network Configuration Summary Table |                  |                                                               |
|-------------------------------------------|------------------|---------------------------------------------------------------|
| Subnet Parameter                          | Calculated Value | Purpose / Notes                                               |
| CIDR Notation                             | /25              | 25 Network Bits, 7 Host Bits.                                 |
| Subnet Mask                               | 255.255.255.128  | Used by devices to identify network boundaries.               |
| Network Address                           | 192.168.10.0     | Identifies the local subnet itself (not assignable to hosts). |
| First Usable IP                           | 192.168.10.1     | Typically assigned to the default gateway router interface.   |

|                           |                       |                                                                   |
|---------------------------|-----------------------|-------------------------------------------------------------------|
| <b>Last Usable IP</b>     | <b>192.168.10.126</b> | The final host address available for a local network node.        |
| <b>Broadcast Address</b>  | <b>192.168.10.127</b> | Used to send data broadcast packets to all devices in the subnet. |
| <b>Total Usable Hosts</b> | <b>126 Addresses</b>  | Calculated via standard formula: $(2^7)-2=126$ .                  |

## Question: 2

Your company wants to split the 10.0.0.0/24 network into four equal subnets for different departments (HR, Sales, Tech, Admin). List the network address, broadcast address, and usable IP range for each subnet.

## Answer: 2

### Subnet Design Strategy

The original network is **10.0.0.0/24**, which provides a total pool of **256 absolute IP addresses** (from 10.0.0.0 to 10.0.0.255).

To divide this space into **four equal subnets**, we split the 256 addresses into 4 blocks of **64 addresses each** ( $256 \div 4 = 64$ ). A block size of 64 requires **6 host bits** ( $2^6 = 64$ ), meaning we borrow 2 bits for the network prefix, shifting the subnet mask from a /24 to a **/26** (Subnet Mask: **255.255.255.192**).

### Department Subnet Allocation Table

| Department Name | Network Address      | First Usable IP | Last Usable IP | Broadcast Address | Total Usable Hosts |
|-----------------|----------------------|-----------------|----------------|-------------------|--------------------|
| 1. HR           | <b>10.0.0.0/26</b>   | 10.0.0.1        | 10.0.0.62      | <b>10.0.0.63</b>  | 62 Devices         |
| 2. Sales        | <b>10.0.0.64/26</b>  | 10.0.0.65       | 10.0.0.126     | <b>10.0.0.127</b> | 62 Devices         |
| 3. Tech         | <b>10.0.0.128/26</b> | 10.0.0.129      | 10.0.0.190     | <b>10.0.0.191</b> | 62 Devices         |
| 4. Admin        | <b>10.0.0.192/26</b> | 10.0.0.193      | 10.0.0.254     | <b>10.0.0.255</b> | 62 Devices         |

## Key Subnetting Breakdown Details

- **Subnet 1 (HR):** Spans from .0 to .63. The address .0 acts as the Network identifier, while .63 functions as the local broadcast marker, leaving 62 available device hosts.
- **Subnet 2 (Sales):** Begins directly at the next available block step .64 and runs up to .127.
- **Subnet 3 (Tech):** Starts precisely at .128 and ends at .191.
- **Subnet 4 (Admin):** Occupies the final block allocation space, starting at .192 and reaching the boundary marker .255.

## Question: 3

A startup needs 60 usable IP addresses for its team. What is the smallest CIDR subnet you can assign, and what are the network address, broadcast address, and usable IP range for this subnet?

## Answer: 3

### Determining the Smallest Subnet Size

- **Requirement:** The team needs **60 usable IP addresses**.
  - **Finding the Power of 2:** We look for the smallest block size ( $2^n$ ) that can accommodate 60 devices after subtracting 2 (for the network and broadcast addresses):
    - $2^5 = 32$  addresses ( $32 - 2 = 30$  usable) → *Too small*.
    - $2^6 = 64$  addresses ( $64 - 2 = 62$  usable) → **Perfect fit**.
  - **Smallest CIDR Subnet:** A block size of 64 requires 6 host bits ( $32 - 6 = 26$ ). Therefore, the smallest subnet you can assign is a **/26 subnet** (Subnet Mask: **255.255.255.192** ).

### Subnet Configuration Breakdown

Using a standard private Class C base network as a starting reference (e.g., 192.168.1.0), the exact subnetting parameters are calculated as follows:

| Subnet Parameter         | Configured Value       | Technical Explanation                                                  |
|--------------------------|------------------------|------------------------------------------------------------------------|
| <b>CIDR Notation</b>     | <b>192.168.1.0/26</b>  | Smallest safe prefix yielding 62 usable device addresses.              |
| <b>Subnet Mask</b>       | <b>255.255.255.192</b> | 26 bits set to network routing, 6 bits left for host tracking.         |
| <b>Network Address</b>   | <b>192.168.1.0</b>     | The absolute first boundary IP identifying the local team subnet.      |
| <b>First Usable IP</b>   | <b>192.168.1.1</b>     | Generally reserved for the department's gateway router interface.      |
| <b>Last Usable IP</b>    | <b>192.168.1.62</b>    | The final host number available for the startup team's computers.      |
| <b>Broadcast Address</b> | <b>192.168.1.63</b>    | The absolute final boundary IP used to broadcast packets to the team.  |
| <b>Total Usable Pool</b> | <b>62 Addresses</b>    | Leaves a safe buffer of <b>2 extra IP spaces</b> for immediate growth. |

## Question: 4

For the public WiFi at a Zomato restaurant, you are given the IP block 172.16.5.0/26. Calculate the subnet mask, network address, broadcast address, and the first and last usable IPs.

**Hint:** /26 gives you 64 addresses per subnet.

## Answer: 4

### Subnet Calculation Breakdown (172.16.5.0/26)

For a public Wi-Fi network at a Zomato restaurant using the **172.16.5.0/26** block, the calculation is based on the /26 prefix. This means the first 26 bits are dedicated to the network, leaving exactly **6 bits for host devices** (32-26=6)

- **1. Subnet Mask:**
  - In binary, the first 26 bits are set to 1 and the remaining 6 bits are 0:  
11111111.11111111.11111111.11000000

- Converting this binary string back to standard dotted-decimal format gives:  
**255.255.255.192**
- **2. Network Address:**
  - This is the absolute first address of the subnet block that identifies the restaurant's public Wi-Fi space.
  - Value: **172.16.5.0**
- **3. Broadcast Address:**
  - As given in the hint, a /26 subnet provides exactly  $2^6 = 64$  total absolute addresses. Spanning from .0 to .63, the last value marks the broadcast boundary.
  - Value: **172.16.5.63**
- **4. First and Last Usable IP Addresses:**
  - Usable host IPs sit directly inside the boundaries of the Network ID and the Broadcast Address.
  - **First Usable IP: 172.16.5.1** (*Typically assigned to the restaurant's primary gateway router*).
  - **Last Usable IP: 172.16.5.62** (*The final available IP address for a customer's phone or laptop*).

**Zomato Wi-Fi Network Configuration Summary Table**

| Subnet Parameter          | Calculated Value       | Purpose & Allocation Notes                                              |
|---------------------------|------------------------|-------------------------------------------------------------------------|
| <b>CIDR Block</b>         | 172.16.5.0/26          | Standard Class B private block customized for small local zones.        |
| <b>Subnet Mask</b>        | <b>255.255.255.192</b> | Masks the local restaurant network segment boundary.                    |
| <b>Network Address</b>    | <b>172.16.5.0</b>      | Identifies this specific Wi-Fi subnet (cannot be assigned to a device). |
| <b>First Usable IP</b>    | <b>172.16.5.1</b>      | Placed on the Wi-Fi router interface acting as the default gateway.     |
| <b>Last Usable IP</b>     | <b>172.16.5.62</b>     | The maximum available host slot for client connection.                  |
| <b>Broadcast Address</b>  | <b>172.16.5.63</b>     | Relays data packets to every active device connected to the Wi-Fi.      |
| <b>Total Usable Hosts</b> | <b>62 Addresses</b>    | Calculated via: $(2^6) - 2 = 62$ total slots for restaurant visitors.   |

## Question: 5

You receive the following scenario: Flipkart's warehouse needs three subnets supporting at least 30 devices each. Using CIDR notation, assign suitable subnets from the 192.168.1.0/24 range, and specify the network address, broadcast address, and usable IP range for each subnet.

## Answer: 5

### Subnet Design Strategy

- **Requirement:** Flipkart's warehouse requires **three subnets**, and each subnet must support **at least 30 usable devices**.
- **Finding the Power of 2:** We need to find the smallest block size ( $2^n$ ) that accommodates 30 devices after subtracting 2 (for the network and broadcast addresses):
  - $2^4 = 16$  addresses ( $16 - 2 = 14$  usable) → *Too small*.
  - $2^5 = 32$  addresses ( $32 - 2 = 30$  usable) → **Perfect fit**.
- **Determining the CIDR Notation:** A block size of 32 total addresses requires 5 host bits ( $32 - 5 = 27$ ). Therefore, each subnet will use a **/27 prefix** (Subnet Mask: **255.255.255.224** ).

### Flipkart Warehouse Subnet Allocation Table

Dividing the base **192.168.1.0/24** range into equal **/27** blocks gives us up to 8 available subnets. The first three subnets assigned for the warehouse layout are detailed below:

| Subnet Allocation    | Network Address | First Usable IP | Last Usable IP | Broadcast Address | Total Usable Hosts |
|----------------------|-----------------|-----------------|----------------|-------------------|--------------------|
| Subnet 1<br>(Zone A) | 192.168.1.0/27  | 192.168.1.1     | 192.168.1.30   | 192.168.1.31      | 30 usable IPs      |
| Subnet 2<br>(Zone B) | 192.168.1.34/27 | 192.168.1.33    | 192.168.1.62   | 192.168.1.63      | 30 usable IPs      |
| Subnet 3<br>(Zone C) | 192.168.1.14/27 | 192.168.1.65    | 192.168.1.94   | 192.168.1.95      | 30 usable IPs      |

## Detailed Network Parameter Breakdown

- **Subnet 1:** Spans from `.0` to `.31`. The address `192.168.1.0` acts as the Network identifier, while `192.168.1.31` functions as the local broadcast marker, providing exactly 30 usable device IP assignments.
- **Subnet 2:** Begins directly at the next available block boundary step `.32` and runs up to `.63`, yielding exactly 30 usable device IP assignments.
- **Subnet 3:** Commences precisely at the next block marker `.64` and reaches the broadcast boundary at `.95`, giving exactly 30 usable device IP assignments.